

VAST Challenge 2019: Disaster at St. Himark!

Mini-Challenge 1: Crowd- sourcing for Situational Awareness

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Figure 1: VAST Challenge 2019: Disaster at St. Himark [[vas19c](#)]

Abstract

This work presents an implementation of the 2019 VAST Challenge: Disaster at St. Himark! Mini-Challenge 1: Crowdsourcing for Situational Awareness. This challenge simulates the aftermath of a disastrous earthquake in the city of St. Himark, where city officials and emergency responders must quickly assess damage across individual neighborhoods and deploy the city's limited resources effectively. Conveniently, St. Himark launched a damage-reporting mobile application for its citizens shortly before the disaster. Thus, this paper introduces an interactive, visual analytics tool that incorporates citizen-reported data and provides a clear overview of the city's conditions to support quick and informed decision-making during an emergency response.

1. Introduction

In the aftermath of an earthquake as well as other natural disasters, emergency services must be able to quickly evaluate the situation on the ground and dispatch their resources effectively. While data sources like seismic readings offer a valuable insight into the shaking intensity and an initial estimate of the damage in different regions, they fail to provide more detail or information about the true conditions. One solution to this problem could be the use of crowd-sourced data, which forms the basis of Mini-Challenge 1 (MC1) of the 2019 VAST Challenge [[vas19b](#)].

The 2019 VAST Challenge introduces a fictional city of

St. Himark, which was hit by a devastating earthquake. Because of the city's limited resources, city officials and emergency services must quickly assess the damage, identify hardest-hit regions, and allocate the resources accordingly. They base the initial response on seismic data; however, more information is needed to understand the actual conditions and coordinate the next steps. Fortunately, a mobile application allowing citizens to report damage across the city was launched shortly before the earthquake, providing the much needed data for the responders. Naturally, these reports require an effective visualization and careful assessment of the reliability of the information.

This paper presents an approach that integrates citizen damage reports into an interactive visual analytics tool, enabling emergency responders to understand the severity and distribution of damage across the city, assess which type of help (e.g. medical aid or power restoration) is most needed in which region, evaluate uncertainty in the data, and monitor how conditions develop over time.

2. Design and Implementation

This section firstly offers an overview of the data officially provided for the challenge, introduces objectives that were defined for the application, mentions some already existing solutions, introduces the visualization components and design choices used for this solution, and finally, shortly presents the technologies used for the implementation.

2.1. Data

The main source of the data for MC1 of VAST Challenge 2019 is a **CSV file** containing individual reports of shaking and damage submitted by citizens via the mobile application over the entire length of the simulation. The system records these reports in 5-minute intervals; however, power outages can cause further delays. The structure of the file is the following [vas19a]:

- time: timestamp in the format *YYYY-MM-DD hh:mm:ss*
- location: id of neighborhood where person reporting is feeling the shaking and/or seeing the damage
- {*shake_intensity*, *sewer_and_water*, *power*, *roads_and_bridges*, *medical*, *buildings*}: reported categorical value of how violent the shaking was/how bad the damage was (1 – low, 10 – highest; missing data allowed)

Other data that were officially provided for the challenge include:

- **Two shake maps** (png files) of two earthquake events from April 6 and April 8 that indicate seismic intensity across the city
- **"About Our City" brochure** – a textual document with various information about St. Himark (e.g. information about neighborhoods, utilities, or emergency services) which can help identify areas with higher priority (i.e. neighborhoods with hospitals)
- **GeoJSON file** containing a vector map of St. Himark

2.2. Objectives

The primary goal of the application is to enable emergency responders to identify areas of concern based on data from citizens and prioritize their response. Taking into account the target users, the overall situation as well as the official tasks of MC1 [vas19b], the following objectives for the application were defined:

- Identify hardest-hit neighborhoods.
- Show uncertainty in the data, helping responders to assess reliability of reports from various neighborhoods.
- Visualize changes over time to monitor how the situation evolves in each region.
- Provide an intuitive and usable interface
- Offer a high-level overview of the whole city as well as detailed analysis of the situation in individual neighborhoods.

Given the urgency of a disaster response, the solution aims to be as intuitive as possible, utilizing well-known visualization components, enabling emergency responders to use the tool immediately and focus on recovery efforts rather than learning new charts.

2.3. Already Existing Solutions

Naturally, because VAST Challenge is a long-running, well-known event, many implementations of the 2019 VAST Challenge already exist. This turned out to be a small challenge on its own, as it was difficult to come up with a new, original idea for a topic that had already been covered by many others. Some existing solutions utilized different well-known visualization components [Kum19,SK21], while others generated more complex approaches [WZCQ19,Adu19].

2.4. Visualization Components

The whole application consists of three main parts:

- **Timeline** – a line chart showing the number of reports over time, allowing users to select a time interval for the data (see the bottom of Fig. 2)
- **City overview** – a set of choropleth maps with several overlay options (see Fig. 2)
- **Detailed region view** – a combination of a heatmap, bar chart and line chart displaying data for a selected neighborhood (see Fig. 3)

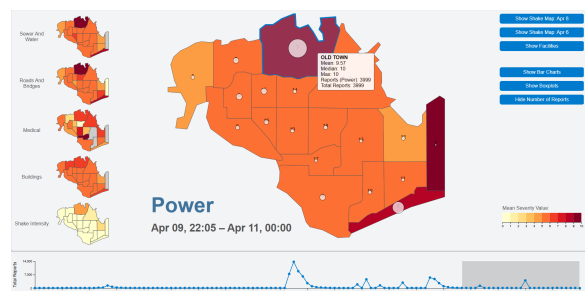


Figure 2: City overview (a set of choropleth maps with "Power" reports selected for the main map), and Timeline. Buttons in the right-side panel allow users to toggle various overlays (e.g. number of reports shown in this figure).

The set of **choropleth maps** — one map for each report



Figure 3: Detailed view of the Palace Hills region over the full time range of the dataset. This view includes a heatmap, grouped bar chart, and line chart, showing the overall situation in the region. While hovering over the charts, all components offer a tooltip which displays more information.

category, with one category always selected for the main map — displays the average reported severity in each region within a selected time interval. This approach was chosen for its intuitiveness and ability to provide a clear overview of conditions across the whole city. Moreover, when hovering over a region, a tooltip showing the average, median, and maximum severity value, and the number of reports in the region, is displayed.

To offer further analysis and comparison of all neighborhoods, users can overlay additional visual elements directly onto the main choropleth map. These include **bar charts** over each neighborhood, showing the distribution of reported severity values (with severity levels on the x-axis and number of reports on the y-axis), **bubbles** (as shown in Figure 2) whose size represents the number of reports in each neighborhood, and **box plots** for the analysis of uncertainty in the data in each neighborhood. Furthermore, users can overlay shake maps and facilities over the main map.

Clicking on a neighborhood in the main choropleth map opens up the detailed view for a selected region, which allows analysis of the situation in the area within a selected time interval. This view includes a line chart, a grouped bar chart and a heatmap, all of which display data of all report categories simultaneously; however, the categories in the bar chart and line chart can be filtered out as needed. This whole view is displayed under the Timeline component, so that the time range selection is easily accessible from both the choropleth view and this detailed view.

The **line chart** (with time on the x-axis and median severity value on the y-axis) indicates how each type of report

changes over time, helping emergency responders determine whether the situation in a neighborhood is worsening or improving. Reports are aggregated into appropriate time intervals to avoid overcrowding.

The **grouped bar chart** displays severity levels on the x-axis and number of reports on the y-axis, providing an overview of severity distribution across all categories. This can help to identify which type of assistance is needed in the region.

Finally, the **heatmap** offers a compact but detailed visualization of all reports. The x-axis shows time in 5-minute intervals, while the y-axis includes all report categories. The color of each rectangle represents the average reported severity for the time and category, enabling responders to quickly identify possibly important events or sudden changes in reports. Gray values indicate that no reports were submitted for the given time and category. Inspiration for the heatmap was taken from an already existing solution [SK21].

2.5. Tools

The application was implemented using **D3.js**[†], a JavaScript library for creating interactive, web-based visualizations. It was chosen mainly for its flexibility in creating custom visualizations, and because of my prior experience with the tool.

[†] <https://d3js.org/>

3. Conclusion and Future Work

This work presented an implementation of VAST Challenge 2019, Mini-Challenge 1: Crowd-sourcing for Situational Awareness, which deals with the impact of a major earthquake in the city of St. Himark. To support effective emergency response efforts, a web-based visual analytics tool was developed, integrating citizen-reported data and providing a comprehensive overview of the conditions on the ground.

Using interactive components such as choropleth maps, bar charts, or heatmaps, the tool enables emergency responders to quickly identify hardest-hit regions, check the reliability of reports, and monitor changes over time. The application focuses on usability and intuitiveness, aiming for a tool that can be used immediately in an urgent situation, without the need to learn new concepts.

Due to limited time, some initially planned features were omitted from the final application and could serve as a direction for future work. One approach involves using a 2D color scale for the choropleth map to show uncertainty – for example, encoding the average severity value with hue, and uncertainty (e.g. computed error margin) with saturation. Similarly, uncertainty could be incorporated to the line chart in the detailed view by adding an area around each line to represent confidence intervals, providing clearer insight into report reliability over time.

Moreover, the application could benefit further optimization by eliminating unnecessary operations and redraws of components, improving responsiveness, and better handling

and visualization of missing values, as I was initially handling them incorrectly and only realized that shortly before the project submission.

References

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